SCIENCE
ENGINEERING
PEOPLE
A BRIGHTER
WORLD

KNOWLEDGE
BUILDS
BETTER
TOMORROWS

COLLEGE OF ENGINEERING & TECHNOLOGY

Department: Basic and Applied Sciences

EBA0201 - PRECALCULUS

$$\frac{d}{dx}[\sin x] = \cos x$$

$$\frac{d}{dx}[\cos x] = -\sin x$$

$$\frac{d}{dx}[e^x] = e^x$$

$$\frac{d}{dx}[\tan x] = \sec^2 x$$

$$\frac{d}{dx}[\ln x] = \frac{1}{x}$$

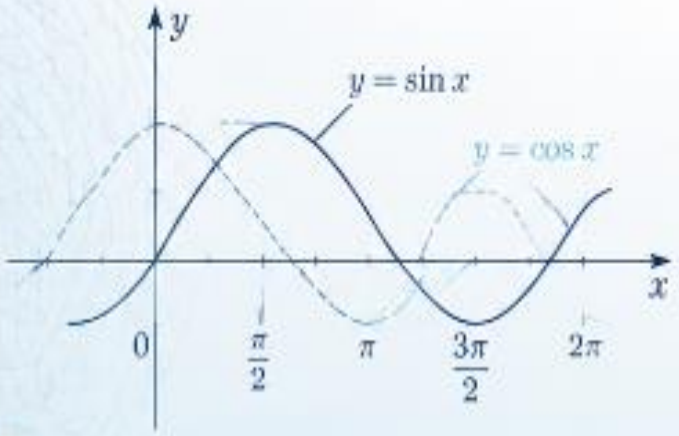
$$\frac{d}{dx}[\log_a x] = \frac{1}{x \ln a}$$

Lecture 3

BASIC RULES OF DIFFERENTIATION

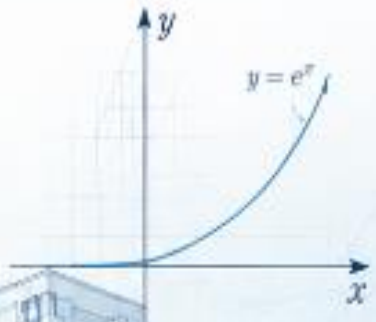
(Trigonometric functions, the exponential function, and the logarithmic function.)

Concept | Method | Worked Examples | Practice



$$\frac{d}{dx}[e^x] = e^x$$

$$\frac{d}{dx}[\ln x] = \frac{1}{x}$$





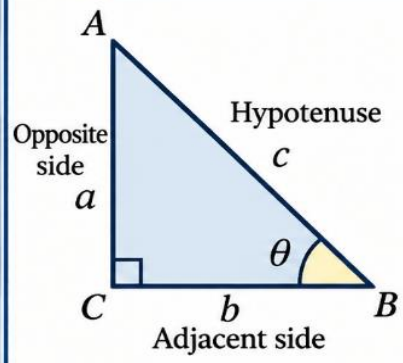
Coordinator:
Dr. Taha Salah El-Din
Dr. Ola Hamada

MATHEMATICS
FOR A BETTER TOMORROW

ANALYZE | MODEL | SOLVE | CREATE

1. Trigonometric Function

Trigonometric Functions Using a Right Triangle



Sine

$$\sin\theta = \frac{\text{opposite side}}{\text{hypotenuse}}$$

$$\sin\theta = \frac{a}{c}$$

Cosine

$$\cos\theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$$

$$\cos\theta = \frac{b}{c}$$

Tangent

$$\tan\theta = \frac{\text{opposite side}}{\text{adjacent side}}$$

$$\tan\theta = \frac{a}{b}$$

Cotangent

$$\cot\theta = \frac{\text{adjacent side}}{\text{opposite side}}$$

$$\cot\theta = \frac{b}{a}$$

Secant

$$\sec\theta = \frac{\text{hypotenuse}}{\text{adjacent side}}$$

$$\sec\theta = \frac{c}{b}$$

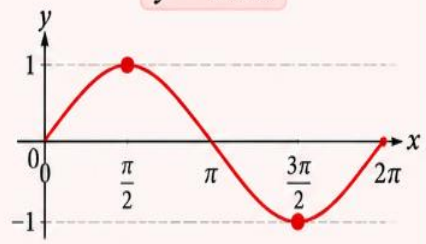
Cosecant

$$\csc\theta = \frac{\text{hypotenuse}}{\text{opposite side}}$$

$$\csc\theta = \frac{c}{a}$$

Sine Function

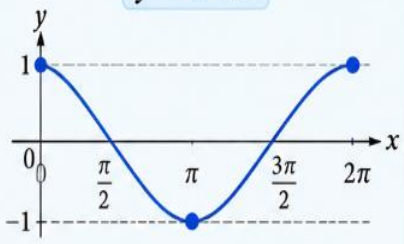
$y = \sin x$



Period : 2π | Range : $[-1, 1]$
Amplitude : 1

Cosine Function

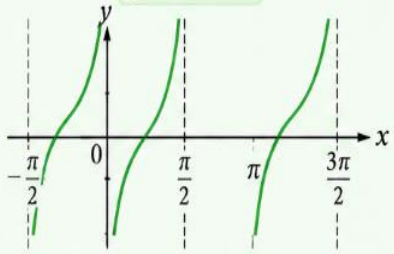
$y = \cos x$



Period : 2π | Range : $[-1, 1]$
Amplitude : 1

Tangent Function

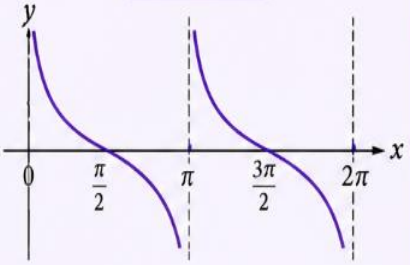
$y = \tan x$



Period : π | Range : $(-\infty, \infty)$
Vertical asymptotes : $x = \frac{\pi}{2} + k\pi$

Cotangent Function

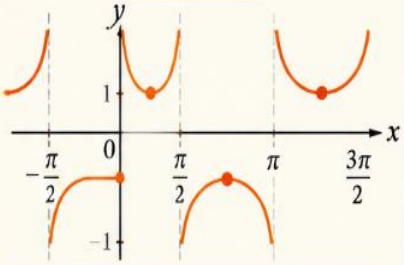
$y = \cot x$



Period : π | Range : $(-\infty, \infty)$
Vertical asymptotes : $x = k\pi$

Secant Function

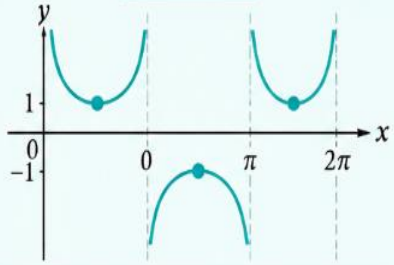
$y = \sec x$



Period : 2π | Range : $(-\infty, -1] \cup [1, \infty)$
Vertical asymptotes : $x = \frac{\pi}{2} + k\pi$

Cosecant Function

$y = \csc x$



Period : 2π | Range : $(-\infty, -1] \cup [1, \infty)$
Vertical asymptotes : $x = k\pi$

2. Trigonometric Derivatives

Six trigonometric rules

$$\frac{d}{dx}(\sin u) = \cos u\ u'$$

$$\frac{d}{dx}(\cos u) = -\sin u\ u'$$

$$\frac{d}{dx}(\tan u) = \sec^2 u\ u'$$

$$\frac{d}{dx}(\cot u) = -\csc^2 u\ u'$$

$$\frac{d}{dx}(\sec u) = \sec u \tan u\ u'$$

$$\frac{d}{dx}(\csc u) = -\csc u \cot u\ u'$$

Example 1

Find $\frac{dy}{dx}$:

$$y = \sin(3x)$$

Solution:

$$y' = 3\cos(3x)$$

Example 2

Find $\frac{dy}{dx}$:

$$y = \tan(x^2 + 1)$$

Solution:

$$y' = 2x\sec^2(x^2 + 1)$$

Example 3

Find $\frac{dy}{dx}$:

$$y = x\tan(x^3 + 3x^2 + 2)$$

Solution:

$$y' = x\sec^2(x^3 + 3x^2 + 2)(3x^2 + 6x) + \tan(x^3 + 3x^2 + 2)$$

Example 4

Find $\frac{dy}{dx}$:

$$y = x^2\sin x + \frac{x}{x-1}$$

Solution: $y' = x^2\cos x + 2x\sin x - \frac{1}{(x-1)^2}$

Example 5

Find $\frac{dy}{dx}$:

$$y = (x^5 + \sin x)\tan^3 x$$

Solution:

$$y' = (5x^4 + \cos x)\tan^3 x + 3(x^5 + \sin x)\tan^2 x\sec^2 x$$

Example 6

Find $\frac{dy}{dx}$:

$$y = \left(\frac{x^2 + \sin x}{x+1}\right)^6$$

Solution:

$$y' = 6\left(\frac{x^2 + \sin x}{x+1}\right)^5 \frac{(x+1)(2x + \cos x) - (x^2 + \sin x)}{(x+1)^2}$$

Example 7

Find $\frac{dy}{dx}$:

$$y = \left(2 - 5\sin\sqrt{x}\right)^3$$

Solution:

$$y' = 3\left(2 - 5\sin\sqrt{x}\right)^2\left[-5\cos\sqrt{x}\left(\frac{1}{2\sqrt{x}}\right)\right] = -\frac{15\cos\sqrt{x}}{2\sqrt{x}}\left(2 - 5\sin\sqrt{x}\right)^2$$

Example 8

Find $\frac{dy}{dx}$:

$$y = x\cos^2(3x-1)$$

Solution: $y' = \cos^2(3x-1) - 6x\cos(3x-1)\sin(3x-1)$

Example 9

Find $\frac{dy}{dx}$:

$$y = \sin(\tan x^2)$$

Solution: $y' = 2x\sec^2(x^2)\cos(\tan x^2)$